INCH-POUND

MIL-A-5815E
AMENDMENT 1
26 September 1991

MILITARY SPECIFICATION

ANTENNA, BLADE, UHF - BAND, GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-A-5815E, dated 18 Mar 1977, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, under 'STANDARDS, MILITARY,' delete: 'MIL-STD-810 - Environmental Test Methods.'

2.1, under 'STANDARDS, MILITARY,' add: 'MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts '

2.2, add:

'INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE STD 149-1979 - Standard Test Procedures for Antennas '

3 2, delete and substitute:

'3.2. First Article. Antennas furnished under this specification shall be products which have been tested in accordance with the first article tests prescribed herein and have been approved by the responsible design activity listed in paragraph 6 3 '

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PAGE 3

3 5, add: "Reinforcing the antenna by wrapping it with tapes or cloths that protrude outside the allowed antenna envelope is prohibited."

3.7, delete and substitute:

"3 7 Impedance. When antennas are tested as specified in 4.7 2, the impedance at the antenna input connector shall be such that the voltage standing wave ratio (VSWR) does not exceed the specified value (see 3.1)."

3.10, delete and substitute:

"3 10 Radiation Pattern. The antenna shall be designed for omnidirectional coverage. When the azimuthal radiation patterns specified in 4 7 5a are taken, the region extending completely around the antenna in azimuth (Ø) shall, at each specified zenith angle (θ), be essentially omnidirectional with no nulls greater than 2 dB down from the maximum When the vertical (great-circle) radiation patterns specified in 4 7.5b are taken, the gain at each zenith angle (θ) shall be no more than 1 dB below the gain of a matched quarter-wave stub antenna mounted in the same location

3 11, delete and substitute·

"3 11 Polarization When antennas are tested as specified in 4.7 6, the axial ratio of the polarization ellipse for the test antenna in each of the specified test directions shall be equal to or greater than the axial ratio in the same direction for a matched quarter-wave stub antenna mounted in the location of the test antenna "

PAGE 4

3.16, delete and substitute·

"3 16 Temperature shock. After the temperature shock test specified in 4 7.11 1, the antenna shall show no physical damage and the impedance and sampling probe impedance shall not exceed the value specified (see 3 1) "

3.17, delete and substitute:

3.17 Barometric pressure (reduced) After the barometric pressure test specified in 4 7 11 2, the antenna shall show no physical damage and the impedance and sampling probe impedance shall not exceed the value specified (see 3 1) "

3 21, change "Humidity" to "Moisture resistance" and change humidity" to "moisture resistance "

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4.2(b), delete and substitute "First Article inspection (see 4 5) "

4.4, last line, change "MIL-STD-810" to "MIL-STD-202."

4.5, change "Qualification inspection" to "First Article inspection" and change "Qualification inspection" to "First Article inspection."

4.5.1, delete and substitute·

"4 5.1 Sample size Four antennas shall be subjected to first article inspection "

4.5 3, delete and substitute:

"4.5.3 Failures. No failures are allowed."

4 5.4, delete

PAGE 6

TABLE II, change title to "First Article inspection " Change "Temperature - Altitude" to "Temperature Shock," change "Leakage (Immersion)" to "Barometric Pressure (Reduced)," and change "Humidity" to "Moisture Resistance."

PAGE 7

4.6 1.2, delete and substitute·

"4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, in the order shown The inspections shall be performed on all antennas in each production lot."

4.6 2, change "Qualification verification inspection" to "Periodic inspection," change "Qualification verification inspection" to "Periodic inspection," and change "qualification verification inspections" to "periodic inspections "

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4 6.2 1 4, last sentence, change "qualifying" to "approving "

TABLE IV, change "Temperature - Altitude" to "Temperature Shock," change "Leakage (Immersion)" to "Barometric Pressure (Reduced)," and change "Humidity" to "Moisture Resistance."

4.7.2, first sentence, delete and substitute: "Impedance measurements shall be made with the antenna mounted at the center of an 8-foot circular ground plane. The long axis of the antenna blade shall be perpendicular to the ground plane. A network analyzer or a slotted-line method may be used for the measurements "

4 7 4, first sentence, delete and substitute· "Attenuation measurements may be made with a network analyzer, or with the following method· A signal source shall be connected through a 50 ohm calibrated RF attenuator to the antenna input connector and the output of the sampling probe shall be connected to a calibrated receiving detector "

PAGE 9

4 7 5, delete and substitute:

4 7 5 Radiation pattern (see 3 10) The radiation pattern shall be measured with the antenna mounted in the center of an eight-foot circular ground plane The measurements shall be made on a continuously recording automatic antenna range. Care shall be taken to avoid errors due to reflections from the earth and from nearby objects (this does not prohibit the use of properly designed reflection ranges) The antenna under test shall be considered as positioned at the origin of the standard right-handed spherical coordinate system shown in Figure 2 of IEEE STD 149-1979, with the plane of the antenna blade in the x-z plane and the long axis of the antenna blade along the z-axis. The leading edge of the blade shall face the direction of increasing x and the top shall face the direction of increasing z The ground plane shall be in the x-y plane of the coordinate system For each sample antenna, polar decibel radiation patterns shall be recorded every 25 MHz from 225 to 400 MHz, as follows:

a Azimuth (conical) patterns ($0° \leq \phi \leq 360°$) for $\theta = 60°$ and $90°$
b Vertical (great-circle) patterns ($0° \leq \theta \leq 90°$) for $\phi = 0°, 90°, 180°$, and $270°$.

Like vertical patterns shall be made for comparison, using the same frequencies and power levels, for a matched quarter-wave stub antenna The stub antenna shall be mounted in the same manner and location and with the same polarization

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as the test antenna Each comparison pattern shall be taken immediately after, and recorded on the same sheet as, the corresponding test antenna pattern There shall be only one test antenna pattern and one comparison pattern per sheet. A separate and distinctive recording trace shall be used such that, on the original and on any copies of the pattern recordings, it is possible to distinguish the test and the comparison antenna patterns. There shall be no change of equipment, equipment settings, personnel, or operating technique that would affect the consistency of the results. All patterns shall be clearly labelled with the sample number, the test frequency, and the angles involved. The labelling of the angles shall be such that, for any pattern, the specific values of θ and ø associated with any measured point can be readily and unambiguously determined."

4.7.6, delete and substitute:

"4 7 6 Polarization (see 3 11) The polarization performance shall be measured using the polarization-pattern method (see IEEE STD 149-1979) The antenna shall be positioned and oriented in the same manner as for the radiation pattern measurements. The test frequency shall be 315 MHz For each test sample, polarization patterns shall be obtained and from them the axial ratio of the polarization ellipse determined, for the following directions (θ, ø):

(60°, 0°), (60°, 90°), (60°, 180°), (60°, 270°)
(90°, 0°), (90°, 90°), (90°, 180°), (90°, 270°)

For comparison, like polarization patterns shall be obtained and the axial ratio of the polarization ellipse determined for a matched quarter-wave stub antenna, using the same frequency and power level The stub antenna shall be mounted in the same manner and location and with the same polarization as the test antenna. Each comparison pattern shall be taken as soon as practicable after the corresponding test antenna pattern such that there is no effect on the consistency of the results Each polarization pattern shall be clearly labelled with the sample number (when applicable) and the direction involved There shall be no change of equipment, equipment settings, personnel, or operating technique that would affect the consistency of the results

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4.7 8, fifth sentence, change "10 minutes" to "60 minutes". Sixth sentence, delete "for a period of 1 hour".

PAGE 11

4.7.11, delete and substitute:

"4 7.11 Environmental tests (see 3 6) Environmental testing shall be in accordance with MIL-STD-202, in the sequence shown herein Following each test, the antennas shall be tested for and must meet the requirements of 3.7 and 3.8 The following details shall apply:

4.7.11.1 Thermal shock (see 3.16) Method 107, test condition C

4.7.11.2 Barometric pressure (reduced) (see 3 17). Method 105, test condition C The antenna shall be mounted on a metal plate by its normal mounting means. Prior to reduction of pressure, the VSWR with the antenna mounted in the chamber shall be measured and recorded. After a minimum of 10 minutes at reduced pressure, the VSWR shall again be measured and recorded The antenna shall be considered as passing the test if the VSWR does not change more than ± 20% from the initial reading The VSWR requirement specified in paragraph 3 7 herein does not apply when the antenna is in the chamber

4.7 11 3 Shock (see 3 18) Method 213, test condition J, normal mounting means

4 7.11 4 Vibration (see 3 19). Method 204, test condition B The antenna shall be mounted on the vibration table by its normal mounting means Prior to vibration, the VSWR with the antenna mounted on the table shall be measured and recorded During vibration, the VSWR shall be monitored to evaluate the effects of the test The antenna shall be considered as passing the test if the VSWR does not change more than ± 20% from the initial reading The VSWR requirement specified in paragraph 3 7 herein does not apply when the antenna is on the vibration table.

4.7 11.5 Sand and dust (see 3 20) Method 110

4.7.11 6 Moisture resistance (see 3 21) Method 106 The antenna shall be mounted on a metal plate by its normal mounting means. Prior to testing, the VSWR with the antenna mounted in the chamber shall be measured and recorded After the test, and while still at high humidity, the VSWR shall again be measured and recorded The antenna shall be considered as passing this portion of the test if the VSWR does not change more than ± 20% from the initial

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reading The VSWR requirement specified in paragraph 3.7 herein does not apply when the antenna is in the chamber. Following the measurement at high humidity, the impedance measurements specified in paragraph 4.7.11 above shall be performed in accordance with paragraphs 3 6.2 and 3.6 3 of Method 106.

4.7.11.7 Salt spray (see 3.22) Method 101, test condition B.

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6 2, add.

"(f) First article sampling, test, and evaluation requirements "

6.3, delete and substitute·

"6.3 Design activity The activity responsible for antennas procured under this document is 2750LOG/ESP, Gentile AFS OH 45444 "

CONCLUDING MATERIAL

Custodians·
Army - ER
Air Force - 85

Review activities·
Army - AV
Air Force - 11, 99
DLA - ES

Preparing activity·
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